

HW2

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P1 3.58

```
long decode2(long x, long y, long z){  
    y -= z;  
    x *= y;  
    tmp = y;  
    tmp <<= 63;  
    tmp >>= 63;  
    return x ^ tmp;  
}
```

P2 3.60

- A. x : %rdi n : %esi result : %rax mask : %rdx
- B. result : 0 mask : 1
- C. 是否为0
- D. 通过 salq %cl, %rdx 修改
- E. 通过 orq %r8, %rax 修改

```

long loop (long x, int n) {
    long result = 0;
    long mask;
    for (mask = 1; mask != 0; mask = mask << n) {
        result |= result & mask;
    }
    return result;
}

```

P3 3.63

从0x3c~0x3x+5 分别对应了跳转表的六个语句，因此

```

long switch_prob (long x, long n) {
    long result = x;
    switch (n) {
        case 60: result = x * 8; break;
        case 61: result = x + 0x4b; break;
        case 62: result = x * 8; break;
        case 63: result = x >> 3; break;
        case 64: result = x << 4 - x;
        case 65: result = x * x;
        default: result = x + 0x4b; break;
    }
    return result;
}

```

P4 3.69

容易看出来，结构体数组占了 $0x120 - 0x8 = 0x118 = 280B$

单个a_struct 占 $(4 + 1) \times 8 = 40B$

因此 $CNT = \frac{280}{40} = 7$

idx 占 0x8B，类型为long， a_struct 占5个8B， 因此其定义为

```
typedef struct{
    long idx;
    long x[4];
} a_struct;
```

P5 3.70

- A. e1.p : 0, e1.y : 8, e1.x : 0, e1.next : 8
- B. 共16个字节
- C. `up->e2.x = *(up->e2.next->p) - up->e2.next->e1.y`